

$$\Omega = \{a, b, c, d\}.$$

$$\mathcal{F}_1 = \{\Omega, \phi, \{a\}\}, \quad \mathcal{F}_2 = \{\Omega, \phi, \{a\}, \{a\}^c\}.$$

$$\mathcal{F}_3 = \mathcal{P}(\Omega), \quad \mathcal{F}_4 = \{\phi, \Omega\}.$$

$$\mathcal{F}_2' = \{\phi, \Omega, \{b\}, \{a, c, d\}\}, \quad \mathcal{F}_2'' = \{\Omega, \phi, \{a, b\}, \{c, d\}\}$$

$\{b\}^c$ $\nwarrow$ $\nearrow$
both alg. (σ -alg.)

Any finite alg is σ -alg.

not an algebra.

$$\mathcal{F}_2''' = \mathcal{F}_2' \cup \mathcal{F}_2'' = \{\phi, \Omega, \{b\}, \{a, c, d\}, \{a, b\}, \{c, d\}\}$$

note $\mathcal{F}_2'''$ has (a), (b), (d') but not (c).

Note: in defn. of σ -alg, (a), (b), (c) & (d), we can drop (c) since (d) $\Rightarrow$ (c).

$$A_1 \cup A_2 \cup \phi \cup \phi \dots = A_1 \cup A_2$$

$$\mathcal{F}_2''' = \mathcal{F}_2' \cap \mathcal{F}_2'' = \{\phi, \Omega\} = \mathcal{F}_4.$$

Note if $\mathcal{F}$ and $\mathcal{G}$ are algebras / σ -algebra

$\mathcal{F} \cap \mathcal{G}$ has to be algebra / σ -alg.

* In general for any collection of alg (or σ -alg)

$$\{\mathcal{F}_\alpha\}_{\alpha \in I}, \quad \mathcal{G} = \bigcap_{\alpha \in I} \mathcal{F}_\alpha \text{ will be an alg (or } \sigma\text{-alg)}$$

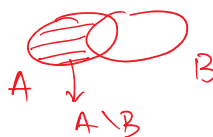
(check),

* If $\mathcal{F}$ is alg (or σ -alg), and $A, B \in \mathcal{F}$ then

$$A \setminus B \in \mathcal{F}.$$

(Check)

$$A \cap B^c$$



$$\Omega \neq \phi$$

$$\mathcal{F}_5 = \{A \subset \Omega : A \text{ or } A^c \text{ is finite}\} \text{ is an alg.}$$

proof: ① show $\Omega \in \mathcal{F}_5$

$$\phi \text{ is finite, } \Omega = \phi^c \rightarrow \Omega \in \mathcal{F}_5$$

② $A \in \mathcal{F}_5$ show $A^c \in \mathcal{F}_5$

$$\text{case 1: } A \text{ finite: } (A^c)^c = A \text{ so } A^c \text{ co-finite}$$

(2) $A \in \mathcal{F}_\Sigma$ show $A^c \in \mathcal{F}_\Sigma$

Case I : A finite : $(A^c)^c = A$ so A^c co-finite
 $A^c \in \mathcal{F}_\Sigma$

Case II : A^c finite (A co-finite)
 so $A \in \mathcal{F}_\Sigma$

(3) $A, B \in \mathcal{F}_\Sigma$ show $A \cup B \in \mathcal{F}_\Sigma$

Case I : A finite, B finite.

$A \cup B$ is finite set so $A \cup B \in \mathcal{F}_\Sigma$

Case II : A co-finite, B finite.

$A \cup B = (A^c \cap B^c)^c \rightarrow$ co-finite
 so $A \cup B \in \mathcal{F}_\Sigma$

$\uparrow$
 finite

Case III : A finite, B co-finite. $\rightarrow$ same pf.

Case IV : A - co-finite, B - co-finite

$\rightarrow$ Case II' : if at least one of A, B co-finite.
 then $A^c \cap B^c$ will be finite
 so $A \cup B = (A^c \cap B^c)^c$ is co-finite
 $\Rightarrow A \cup B \in \mathcal{F}_\Sigma$

$$\Omega \neq \emptyset, \mathcal{A} \subset \mathcal{P}(\Omega), \sigma(\mathcal{A}) = \bigcap \{ \mathcal{G} \mid \mathcal{A} \subset \mathcal{G}, \mathcal{G} \text{ } \sigma\text{-alg} \}$$

note that $\sigma(\mathcal{A})$ is the "smallest" σ -alg cont $\mathcal{A}$.

Since if $\mathcal{F} \supseteq \mathcal{A}$ and $\mathcal{F}$ σ -alg, (i.e. $\mathcal{F} \in \mathcal{I}(\mathcal{A})$)

then $\sigma(\mathcal{A}) \subset \mathcal{F}$.

Note: * If $\mathcal{A}$ is a σ -alg $\sigma(\mathcal{A}) = \mathcal{A}$
 * $\sigma(\sigma(\mathcal{A})) = \sigma(\mathcal{A})$ for any $\mathcal{A}$.

* If $\mathcal{A} \subset \mathcal{B}$, $\sigma(\mathcal{A}) \subset \sigma(\mathcal{B})$
 (check) $\Omega = \{a, b, c, d\}$
 $\mathcal{A} = \{\emptyset, \{a\}, \{a, b\}, \Omega\}$
 $\mathcal{B} = \{\emptyset, \{a, b\}, \{a, c\}, \Omega\}$

* 1 | ... (check)

$$\Omega = \{a, b, c, d\}$$

$$\mathcal{F}_2 = \{\Omega, \{a\}, \{a\}^c, \emptyset\}$$

$$\mathcal{F}_1 = \{\emptyset, \{a\}, \Omega\}$$

show: $\sigma(\mathcal{F}_1) = \mathcal{F}_2$.

$$\mathcal{F}_2 \text{ } \sigma\text{-alg} \supset \mathcal{F}_1$$

$$\sigma(\mathcal{F}_1) \subset \mathcal{F}_2 \quad \checkmark$$

Need to show $\bigcap \mathcal{G} = \sigma(\mathcal{F}_1) \supset \mathcal{F}_2$.

take $\mathcal{G} \text{ } \sigma\text{-alg}$, $\mathcal{F}_1 \subset \mathcal{G}$,

$$\{a\} \in \mathcal{F}_1 \subset \mathcal{G}$$

$$\{a\}^c \in \mathcal{G} \text{ (since } \mathcal{G} \text{ } \sigma\text{-alg)}$$

Note $\emptyset, \{a\}, \Omega \in \mathcal{F}_1 \subset \mathcal{G}$.
we showed $\{a\}^c \in \mathcal{G}$. } so $\mathcal{F}_2 = \{\emptyset, \Omega, \{a\}, \{a\}^c\} \subset \mathcal{G}$.

Borel σ -algebra.

$$a, b \in \mathbb{R}$$

$$(a, b) = (-\infty, b) \setminus (-\infty, a] = (-\infty, b) \cap \left((-\infty, a] \right)^c$$

$$\downarrow$$

$$\in \mathcal{O}_2$$



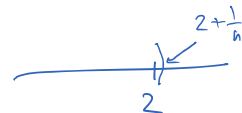
$$\mathcal{O}_2 = \{(a, b)\}, \mathcal{O}_1 = \{(-\infty, b)\}$$

$$= (-\infty, b) \cap \left(\bigcap_{n \geq 1} (-\infty, a + \frac{1}{n}) \right)^c$$

$$\in \sigma(\mathcal{O}_1)$$

$$\mathcal{O}_2 \subset \sigma(\mathcal{O}_1)$$

$$\sigma(\mathcal{O}_2) \subset \sigma(\mathcal{O}_1)$$



$$\bigcap (-\infty, 2 + \frac{1}{n}) = (-\infty, 2]$$

examples of measures:

$$\Omega = \{a, b, c, d\}, \mathcal{F}_2 = \{\emptyset, \{a\}, \{b, c, d\}, \Omega\}$$

$$\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} a & b & c & d \end{bmatrix} \Omega$$

$$\mu_1(\emptyset) = 0$$

$$\mu_1(\{a, b, c\}) = 3$$

$$\mu_1 \quad \begin{matrix} \downarrow & \downarrow & \downarrow & \downarrow \end{matrix} \quad \begin{matrix} 0 & 1 & 3 & 4 \end{matrix}$$

$$\mu_2 \quad \begin{matrix} \downarrow & \downarrow & \downarrow & \downarrow \end{matrix} \quad \begin{matrix} 0 & 1/4 & 3/4 & 1 \end{matrix}$$

$$\mu_3 \quad \begin{matrix} \downarrow & \downarrow & \downarrow & \downarrow \end{matrix} \quad \begin{matrix} 0 & 2 & 4 & 6 \end{matrix}$$

$$\mu_4 \quad \begin{matrix} \downarrow & \downarrow & \downarrow & \downarrow \end{matrix} \quad \begin{matrix} 0 & 2 & \infty & \infty \end{matrix}$$

$$\mu_3(a) = \text{not defined}$$

$$\mu_3(\{a\}) = 2$$

$$\mu_3(\{\emptyset\}) = \text{not defined.}$$

$$\mu_5 \quad 0 \quad \infty \quad \infty \quad \infty$$

Fact 0 Let $\Omega \neq \emptyset$, $\mathcal{F}$ alg (or σ -alg), μ meas., $A \subset B$, $A, B \in \mathcal{F}$.

$$(i) \quad \mu(B \setminus A) + \mu(A) = \mu(B)$$

$$(ii) \quad \mu(B \setminus A) = \mu(B) - \mu(A), \text{ if } \mu(A) < \infty$$

$$(iii) \quad \mu(A) \leq \mu(B)$$



Pf: $B \setminus A \in \mathcal{F}$ and $(B \setminus A) \cup A = B$

$$\mu(B) = \mu((B \setminus A) \cup A) = \mu(B \setminus A) + \mu(A)$$

by (c) of definition.

that proves (i)

(ii) follows from (i)

$$(iii) \quad \text{from (i)} \quad \mu(B) = \underbrace{\mu(B \setminus A)}_{\geq 0} + \mu(A) \quad \text{by (a) of defn.}$$

$$\Rightarrow \mu(B) \geq \mu(A)$$

Trivial Fact 1

If μ is a meas on σ -alg $\mathcal{F}$ and $\mathcal{G} \subseteq \mathcal{F}$ is a σ -alg, then μ is a meas. on $\mathcal{G}$. (check).

Leb.-Stieljes meas on $\mathbb{R}$. ($\Omega = \mathbb{R}$)

Start with $\mathcal{C} = \{(a, b] : a < b\} \leftarrow \text{"semi-alg"}$

take a $\uparrow$ "cdf-like" fn. F

$$\text{eg: } F(x) = x.$$

$$F(x) = \int_{-\infty}^x \frac{1}{\sqrt{2\pi}} e^{-u^2/2} du \equiv \Phi(x) \quad \text{Gaussian cdf.}$$

$$F(x) = \sum_{\substack{i \leq x \\ i \in \{0,1,\dots,n\}}} \binom{n}{i} p^i (1-p)^{n-i} \quad \leftarrow \text{Binomial cdf.}$$

define $\mu_F((a, b]) = F(b) - F(a)$, for $(a, b] \in \mathcal{C}$.

will define a measure on "semi-alg" $\mathcal{C}$.
(will show)

next section proves That one can extend this to a measure on $\sigma(\mathcal{C}) = \mathcal{B}(\mathbb{R})$.

and if μ is finite (eg prob.) or σ -finite
This extension is unique!

$\mathcal{C}$'s extn. thm:

define $\mathcal{A}(\mathcal{C}) = \bigcap_{\mathcal{A} \text{ alg} \supset \mathcal{C}} \mathcal{A} \leftarrow \text{smallest alg cont } \mathcal{C}$.

check $\mathcal{A}(\mathcal{C}) = \widetilde{\mathcal{A}}(\mathcal{C}) \equiv \left\{ \bigcup_{i=1}^k B_i : k \geq 1, B_i \in \mathcal{C} \right\}$

also check: $\widetilde{\mathcal{A}}(\mathcal{C}) = \left\{ \bigcup_{j=1}^n C_j : n \geq 1, C_j \in \mathcal{C}, C_i \cap C_j = \emptyset \right\}$

$A, B \in \mathcal{C}$.

$$A \cup B = (A \cap B^c) \cup_{d_j} (A \cap B) \cup_{d_j} (A^c \cap B)$$



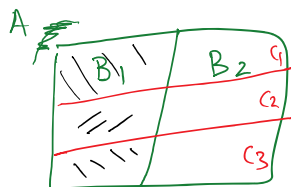
$$= [A \cap (\bigcup_{i=1}^{k_1} B_i)] \cup \dots \cup (\dots$$

$$= [(A \cap B_1) \cup_{d_j} (A \cap B_2) \dots] \cup_{d_j} \dots \cup_{d_j} \dots$$

prop 1.3.1

(i) (idea!) $A = \bigcup_{d_j} B_1 \cup B_2 = \bigcup_{d_j} C_1 \cup C_2 \cup C_3$.

$$\mu(A) : \quad \mu(B_1) + \mu(B_2) \stackrel{?}{=} \mu(C_1) + \mu(C_2) + \mu(C_3)$$

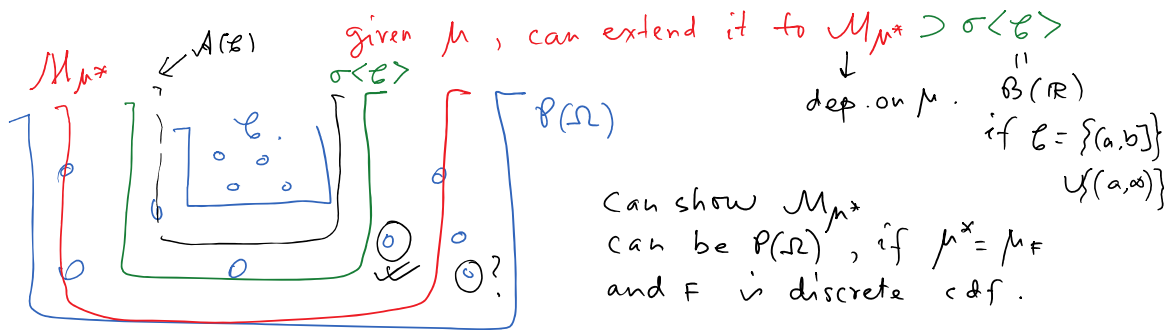


$$\begin{aligned} \mu(B_1) &= \mu(B_1 \cap A) = \mu(B_1 \cap (C_1 \cup C_2 \cup C_3)) \\ &= \mu((B_1 \cap C_1) \cup_{d_j} (B_1 \cap C_2) \cup_{d_j} (B_1 \cap C_3)) \\ &= \mu(B_1 \cap C_1) + \mu(B_1 \cap C_2) + \mu(B_1 \cap C_3) \end{aligned}$$

$$\begin{aligned} \mu(B_1) &= \mu(B_1 \cap C_1) + \mu(B_1 \cap C_2) + \mu(B_1 \cap C_3) \\ \mu(B_2) &= \mu(B_2 \cap C_1) + \mu(B_2 \cap C_2) + \mu(B_2 \cap C_3) \end{aligned}$$

$$\text{||ly } \mu(B_2) = \underbrace{\mu(B_2 \cap C_1)} + \mu(B_2 \cap C_2) + \mu(B_2 \cap C_3)$$

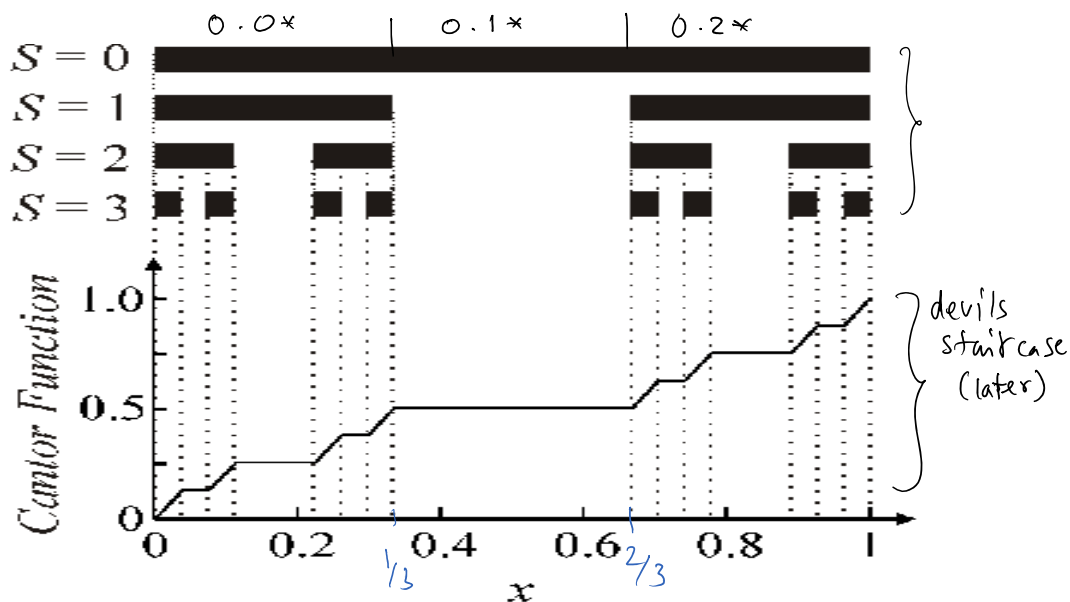
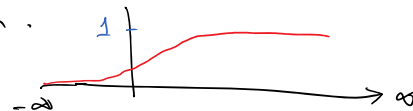
$$\mu(B_1) + \mu(B_2) = \mu(C_1) + \mu(C_2) + \mu(C_3)$$



cdf: cumulative distr. function.

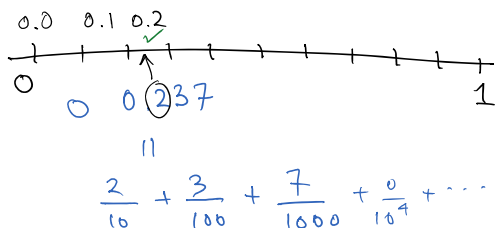
F right-cont

$$F(\infty) = 1, F(-\infty) = 0, F \geq 0$$

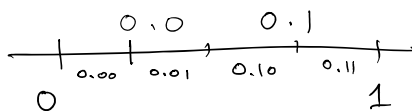


set for $S=1$, $S=2$, and $S=3$. The structure for

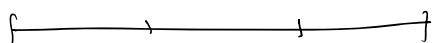
decimal



binary



ternary



Leb. meas. trans / refl. inv. (property (iv))

Show: $m(E+c) = m(E)$, $c \in \mathbb{R}$.

Pf. define $\mu_c(E) = m(E+c)$, $\forall E \in \mathcal{B}(\mathbb{R})$.

— it is a valid meas.

and $\mu_c(E) = m(E)$ $\forall E \in \mathcal{C} = \{(a,b]\} \cup \{(a,\infty)\}$

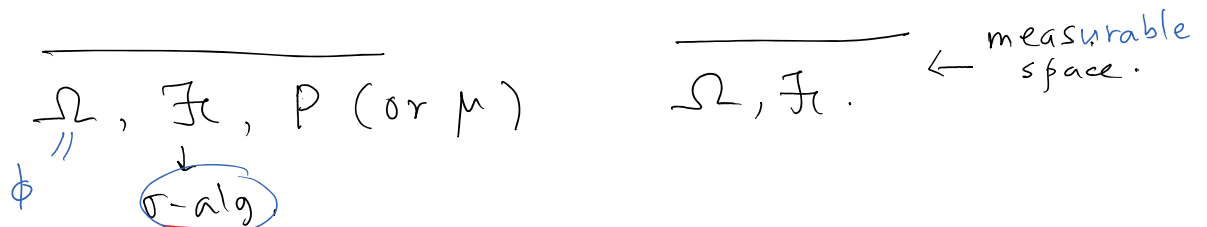
So by uniqueness thm $\mu_c(E) = m(E) \forall E \in \sigma(\mathcal{C}) = \mathcal{B}(\mathbb{R})$
(T.1.2.4 + T.1.3.6.)

$$\begin{aligned} \mu_c((a,b]) &= m((a,b] + c) = m((a+c, b+c]) \\ &= (b+c) - (a+c) = b-a = m((a,b]) \end{aligned}$$

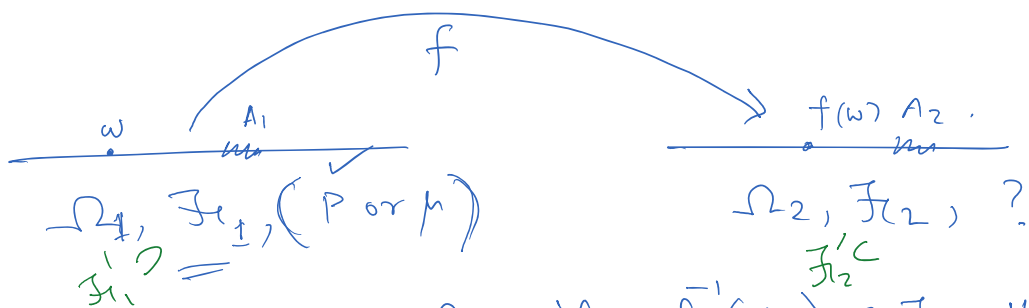
Ch. 2: Random Variables, Expectations etc.

Ch. 1 $\rightarrow$ prob. space (measure space)

Exam 1
next friday
(— Ch. 2.1)
Bring paper



Ch. 2.1

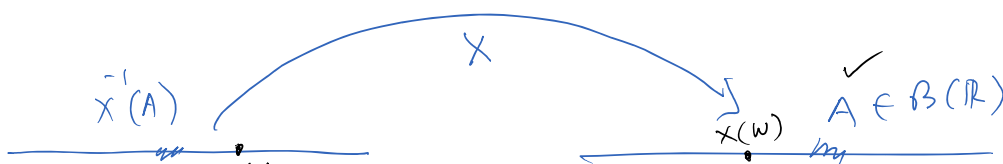


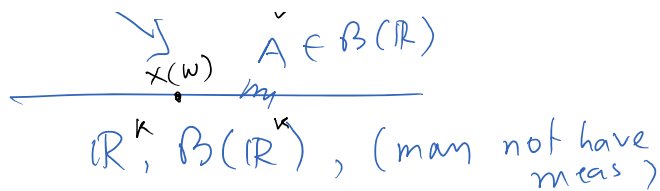
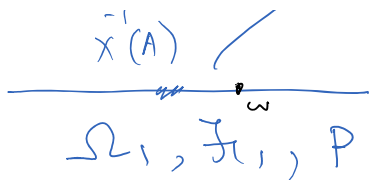
d.2.1.3

f is called measurable $f_{\mathcal{F}_1}$ if $f^{-1}(A_2) \in \mathcal{F}_1 \forall A_2 \in \mathcal{F}_2$

If $\Omega_1, \mathcal{F}_1$ has a prob. P defined on it, $\Omega_2 = \mathbb{R}^k$
 $\mathcal{F}_2 = \mathcal{B}(\mathbb{R})^k$.
then $\#$ a meas. f is called a r.v.

(usually denoted by X .) $X \rightarrow (X_1, \dots, X_k)$ (random vector)





$$X : \Omega_1 \rightarrow \mathbb{R}.$$

can talk about $\rightarrow P(X^{-1}(A)) = P(\{\omega \in \Omega_1 : X(\omega) \in A\})$

$\equiv P(X \in A) \leftarrow \text{Short hand}$

ex! $\Omega_1 = \{HH, HT, TH, TT\}$, $\mathcal{F}_1 = \mathcal{P}(\Omega_1)$

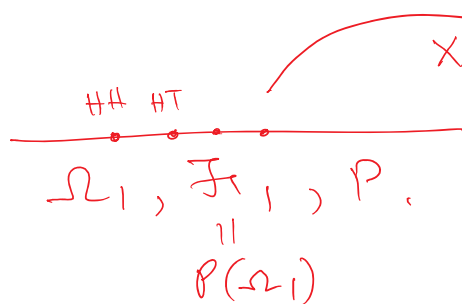
$P(\{HH\}) = p^2$, $P(\{HT\}) = pq$, $P(\{TH\}) = qp$, $P(\{TT\}) = q^2$

where $p+q=1$, $0 \leq p, q \leq 1$, (check: $P(\Omega) = 1$)

$A = \{HH, TT\}$.

$P(A) = p^2 + q^2$.

$X : \begin{matrix} HH \rightarrow 2 \\ HT \rightarrow 1 \\ TH \rightarrow 1 \\ TT \rightarrow 0 \end{matrix}$ } can check easily.



$\Omega_2 = \mathbb{R}$, $\mathcal{F}_2 = \mathcal{B}(\mathbb{R})$

Since $X^{-1}(A)$ is a subset of Ω_1 and $\mathcal{F}_1 = \mathcal{P}(\Omega_1)$

$P(X=1) = P(X \in \{1\}) \leftarrow P_X(\{1\})$

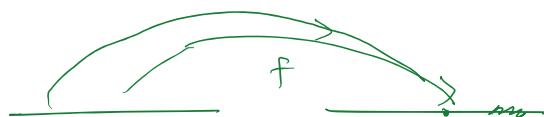
$= P(X^{-1}(\{1\}))$

$= P(\{HT, TH\}) = 2pq$

ex: $f : (\Omega_1, \mathcal{F}_1) \xrightarrow{\text{meas.}} (\Omega_2, \mathcal{F}_2)$

$f(\omega) = c \in \Omega_2 \quad \forall \omega \in \Omega_1$

take any $A \in \mathcal{F}_2$.



$$f^{-1}(A) = \begin{cases} \Omega_1 & , \text{ if } c \in A \\ \emptyset & , \text{ if } c \notin A \end{cases} \quad \Omega_1, \mathcal{F}_1 \quad \Omega_2^c, \mathcal{F}_2$$

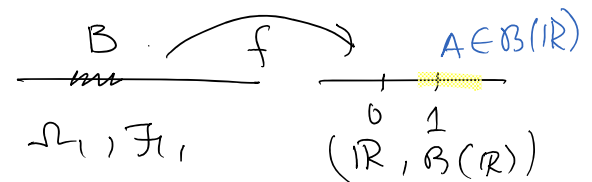
$\in \mathcal{F}_1$ (always since $\mathcal{F}_1$ is σ -alg of Ω_1)

So const. fns are always meas. (any $\Omega_1, \mathcal{F}_1, \dots$)

ex: $f = 1_B$, $B \subset \Omega_1$. ($B \in \mathcal{F}_1$)

$A \in \mathcal{B}(\mathbb{R})$

$$f^{-1}(A) = \begin{cases} \Omega_1 & , \text{ if } 0, 1 \in A \\ B^c & , \text{ if } 0 \in A, 1 \notin A \\ B & , \text{ if } 0 \notin A, 1 \in A \\ \emptyset & , \text{ if } 0, 1 \notin A \end{cases}$$

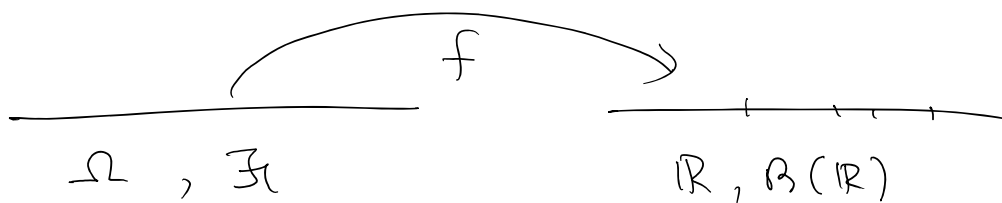


So if $B \in \mathcal{F}_1$, then 1_B is meas.

ex: r.v./meas. fns provide a "partition" of Ω , $\mathcal{F}_1$ needs to be a "finer" partition than that

$$\Omega = \{a, b, c, d\}. \quad \mathcal{F}_1 (= \mathcal{F}_2) = \{\Omega, \{a\}, \{a\}^c, \emptyset\}.$$

$\downarrow$ $\{b, c, d\}$
 incl. 1



$$f : \begin{array}{ll} a & \longrightarrow 5 \\ b & \longrightarrow 17 \\ c & \longrightarrow 10 \\ d & \longrightarrow 100 \end{array}$$

Is it meas? No.
 $f^{-1}(\{10\}) = \{c\} \notin \mathcal{F}_1$

$$\in \mathcal{B}(\mathbb{R})$$

can check: g is meas if

$$g: \begin{cases} a \rightarrow 5 \\ b \rightarrow 17 \\ c \rightarrow 17 \\ d \rightarrow 17 \end{cases} \quad \left. \vphantom{\begin{matrix} a \\ b \\ c \\ d \end{matrix}} \right\} \text{check.}$$

Usually $(\Omega_2, \mathcal{F}_2)$ will be $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$
(or $\mathbb{R}^k, \mathcal{B}(\mathbb{R}^k)$)

$$\begin{array}{ccc} & \xrightarrow{f} & A \in \mathcal{F}_2 \\ \Omega_1, \mathcal{F}_1 & & \end{array}$$

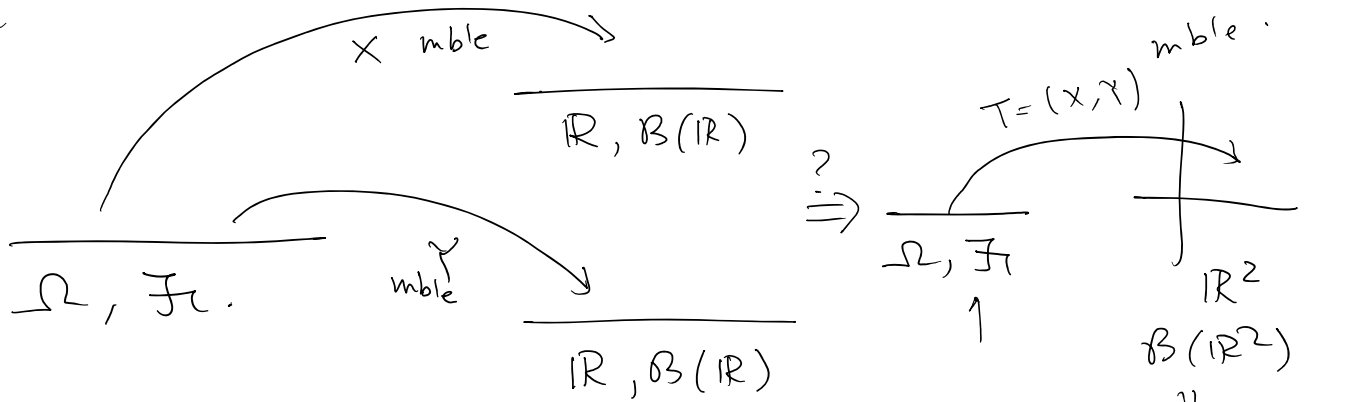
$\Omega_2, \mathcal{F}_2 = \sigma\langle \mathcal{A} \rangle$ for any collection $\mathcal{A}$ of Ω_2 .
In this case enough to verify

$$\underline{f^{-1}(A) \in \mathcal{F}_1} \quad \forall A \in \underline{\mathcal{A}}. \quad \text{prop 2.1.1.}$$

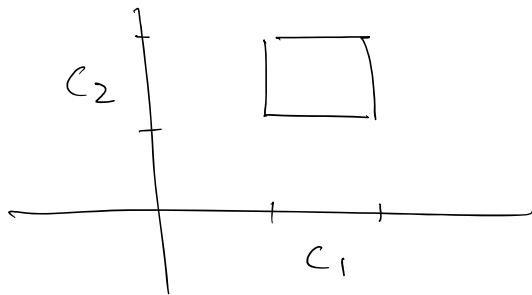
$$\mathcal{B}(\mathbb{R}) = \sigma\langle \mathcal{O} \rangle = \sigma\langle \mathcal{O}_i \rangle \quad i=1,2,\dots$$

measurable?

- ✓ constant f's
- ✓ indicator of measurable sets $\begin{matrix} 1_B \\ B \in \mathcal{F}_1 \end{matrix}$
- ✓ composition of meas. f's.
- ✓ cont. f's
 - (f_1, f_2) mble $f_1 \pm f_2$, $f \cdot f_2$ etc mble (p. 2.1.3)
 - $\max, \min, \{f=g\} \in \mathcal{F}$.
 - $\limsup, \liminf$ etc. (p. 2.1.5)



$$C = C_1 \times C_2$$



$$\begin{aligned} T^{-1}(C) &= \{ \omega : (x, y)(\omega) \in C_1 \times C_2 \} \\ &= \{ \omega : x(\omega) \in C_1, y(\omega) \in C_2 \} \\ &= \{ \omega : x(\omega) \in C_1 \} \cap \{ \omega : y(\omega) \in C_2 \} \\ &= \underbrace{x^{-1}(C_1)}_{\in \mathcal{F}} \cap \underbrace{y^{-1}(C_2)}_{\in \mathcal{F}} \\ &\in \mathcal{F}. \end{aligned}$$

* $\max(f, g), \min(f, g)$, mble. if f, g mble
 $f, g : \Omega \rightarrow \mathbb{R}$

$$\{ \omega : \min(f, g) \in C \} \stackrel{?}{\in} \mathcal{F}$$

$C \in \mathcal{C}_i$
 $[a, \infty)$

$$\{ \omega : \min(f, g) \geq a \} \stackrel{?}{\in} \mathcal{F}.$$

$$\{ f \geq a \} \cap \{ g \geq a \}$$

$\in \mathcal{F} \quad \in \mathcal{F} \quad \checkmark$

$$\min(x, y) = -\max(-x, -y)$$

$$- \min(-x, -y) = \max(x, y), \quad x, y \in \mathbb{R}.$$

$$x \quad A = \{f = g\} \stackrel{?}{\in} \mathcal{F}.$$

$$\stackrel{11}{(f-g)^{-1}} \{0\} \quad \text{Yes!}$$

$$\begin{array}{ccc} & \text{X mble.} & \\ \hline \Omega, \mathcal{F}, & & \mathbb{R}, \mathcal{B}(\mathbb{R}) \end{array}$$

$$\mathcal{G} = \{x^{-1}(B) : B \in \mathcal{B}(\mathbb{R})\} \rightarrow \text{is a } \sigma\text{-alg. (check)}$$

$$x \text{ mble} \iff \mathcal{G} \subset \mathcal{F}. \quad \text{Notation: } \mathcal{G} \equiv \sigma\langle X \rangle.$$

$$\begin{array}{ccc} & T=(X,Y) & \\ \hline \Omega, \mathcal{F} & & \mathbb{R}^2, \mathcal{B}(\mathbb{R}^2) \end{array}$$

$$\sigma\langle X, Y \rangle = \sigma\left\{ \underbrace{x^{-1}(B)}_{\uparrow} : B \in \mathcal{B}(\mathbb{R}) \right\} \cup \{y^{-1}(C) : C \in \mathcal{B}(\mathbb{R})\}$$

↓
smallest σ -alg that makes both x, y mble.

2.2.

$T \langle \mathcal{F}_1, \mathcal{F}_2 \rangle$ -mble.

$$\begin{array}{ccc} & T^{-1}(A) \in \mathcal{F}_1 & \\ \hline \Omega_1, \mathcal{F}_1, \mu & & \Omega_2, \mathcal{F}_2 \end{array} \quad \begin{array}{c} \checkmark \\ A \in \mathcal{F}_2. \end{array}$$

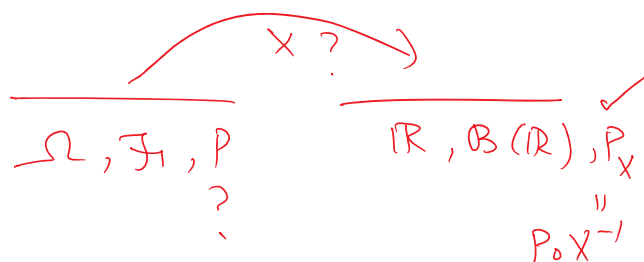
$\mu_T(A) = \mu(T^{-1}(A))$ is a measure on $(\Omega_2, \mathcal{F}_2)$

if μ is prob., call it P , then T is r.v (denote by X)
and $(\Omega_2, \mathcal{F}_2) = (\mathbb{R}, \mathcal{B}(\mathbb{R}))$

$P_X(A) = P(X^{-1}(A))$ is the distr. of r.v X on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$

$$= P(\{\omega : X(\omega) \in A\})$$

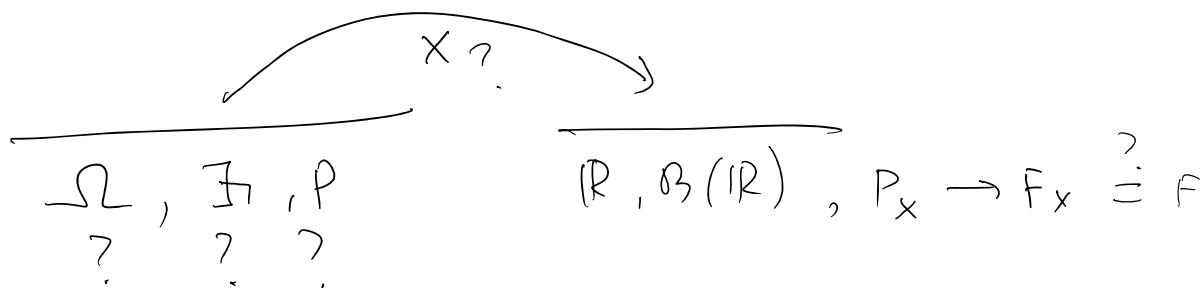
$$= P(X \in A)$$



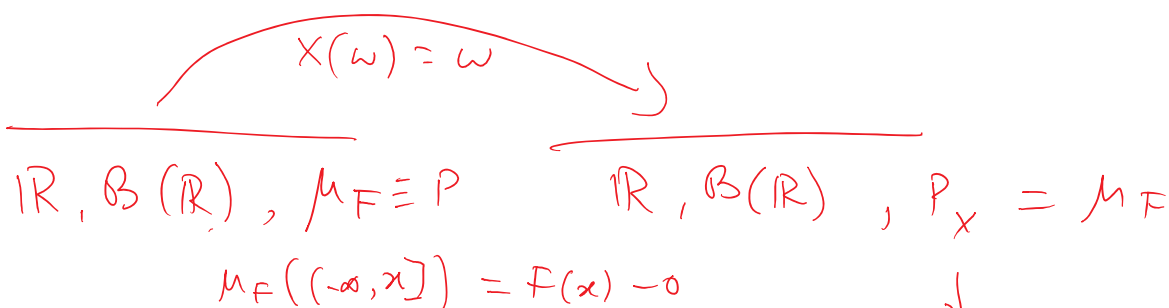
$$P(X \geq 1.2) = P_X((1.2, \infty))$$

$$F: \mathbb{R} \rightarrow \mathbb{R}$$

Given a cdf function F , show that $\exists (\Omega, \mathcal{F}, P)$
and a r.v X on it s.t $F_X = F$.



except for few spl. cases (like Binomial), There
are not many meaningful ways of doing this.



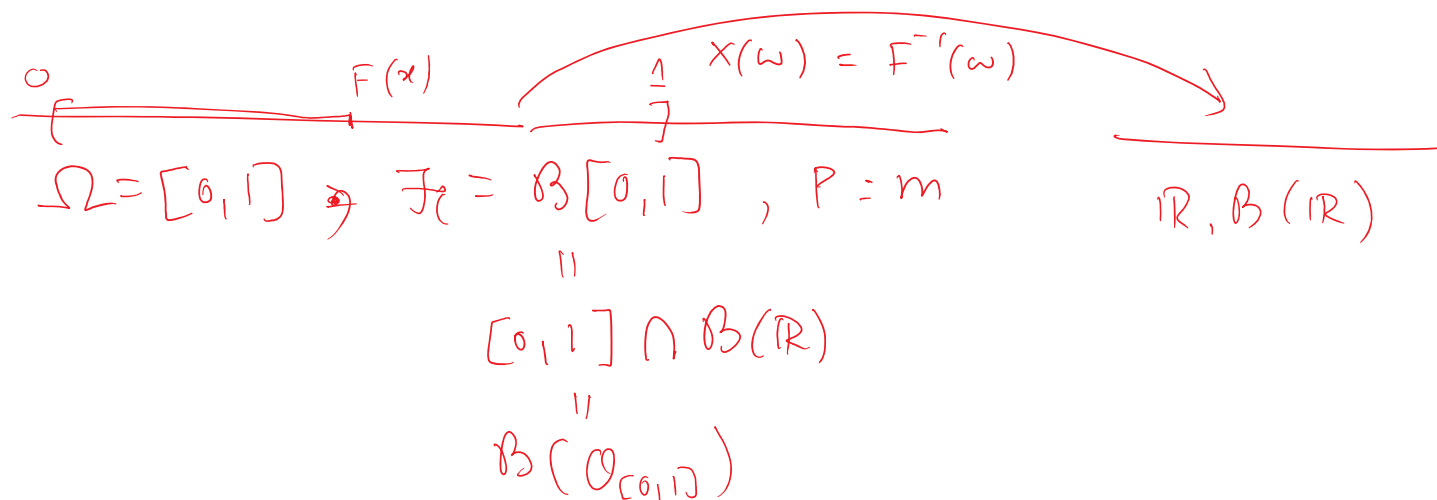
$$\mu_F((-\infty, x]) = F(x) - 0$$

$$P(X \in (a, b)) = \underline{\underline{P_X((a, b))}}$$

$$\begin{aligned} \downarrow \\ F_X(x) &= P_X((-\infty, x]) \\ &= \mu_F((-\infty, x]) \\ &= F(x) \end{aligned}$$

There are other constructions too :

assume F is invertible (i.e. F^{-1} exists)



$$\begin{aligned} F_X(x) &= P(X \leq x) = P(\{\omega : X(\omega) \leq x\}) \\ &= P(\{\omega : F^{-1}(\omega) \leq x\}) \\ &= P(\{\omega : \omega \leq F(x)\}) \\ &= m((0, F(x)]) = F(x) \end{aligned}$$

if F^{-1} does not exist, can use left/right inverse
(Problem 2.15(c))